

ACTIVITY 6 — LEARNER WORKSHEET**Data Collection, KPIs, Check Sheets, and Basic Metrics**

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Lab	45 min	MEASURE	A4

Name	
Date	

What you are doing

Replace anecdote with evidence. Define KPIs with unambiguous operational definitions, design a check sheet, plan the sampling approach, and understand which data type you are collecting — because the data type drives the tool.

What you will produce: A data collection plan with operational definitions, a working check sheet and a sampling plan with the bias risk named.

Tools and techniques: KPI definition, operational definitions, check sheets, sampling, data types

The scenario

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder**data/ — the mock data you analyse:**

- daily-summary.csv — 10 rows (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- service-desk-tickets.csv — 400 rows (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)

templates/ — the worksheets you complete:

- check-sheet.csv (date, ticket_id, ticket_type, assignment_time_min, defect_category, rework_required, notes)
- data-collection-plan.csv (kpi, operational_definition, data_type, data_source, frequency, who_collects, bias_risk)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

data/service-desk-tickets.csv is the full two-week extract (400 tickets).

Your steps

1. Classify your data: continuous vs discrete, nominal vs ordinal — and why it matters.
2. Define KPIs with operational definitions, data source and collection frequency.
3. Design a check sheet capturing date, ticket ID, type, assignment time and defect category.
4. Define mutually exclusive defect categories so every observation lands in exactly one.
5. Plan sampling: how many, how often, by whom — and identify possible bias.
6. State which KPI best reflects the customer pain point from your VOC work.

Your working space

Complete every field. You may keep this worksheet and refer to it during the open-book assessment.

Step 1 — Classify your data: continuous vs discrete, nominal vs ordinal — and why it matters.

Step 2 — Define KPIs with operational definitions, data source and collection frequency.

Step 3 — Design a check sheet capturing date, ticket ID, type, assignment time and defect cate...

Step 4 — Define mutually exclusive defect categories so every observation lands in exactly one.

Step 5 — Plan sampling: how many, how often, by whom — and identify possible bias.

Step 6 — State which KPI best reflects the customer pain point from your VOC work.

Answers to the discussion prompts

1. Would two different people record the same value from your operational definition?

2. Which KPI best reflects the customer pain from your VOC work?

3. Where could bias enter your sampling, and who would introduce it?

Reflect & discuss

What is measured at your workplace that nobody has operationally defined?

Who collects that data, and what incentive do they have to shade it?

Check your work

Two different people reading your operational definition would record the same value for the same event.

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